

# Equilibria

## A Level Chemistry

### Conjugate acids and bases

When a Brønsted acid loses an  $\text{H}^+$ , what is left is its **conjugate base** 共轭碱. When a base gains an  $\text{H}^+$ , it becomes its **conjugate acid** 共轭酸. The two species that differ by just one  $\text{H}^+$  form a **conjugate acid–base pair** 共轭酸碱对.

For example, in  $\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$ , the pair is  $\text{CH}_3\text{COOH}$  (acid) and  $\text{CH}_3\text{COO}^-$  (its conjugate base).

### pH and the equilibrium constants

- **pH** measures acidity:  $\text{pH} = -\log[\text{H}^+]$ , so  $[\text{H}^+] = 10^{-\text{pH}}$ .
- the **acid dissociation constant** 酸解离常数 of a weak acid HA is  $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$ , and  $\text{p}K_a = -\log K_a$ . A larger  $K_a$  (smaller  $\text{p}K_a$ ) means a stronger acid.
- the **ionic product of water** 水的离子积 is  $K_w = [\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$  at 298 K.

### Calculating pH

- **strong acid** 强酸: fully ionised, so  $[\text{H}^+]$  equals the acid concentration; then take  $-\log$ .
- **strong alkali** 强碱: find  $[\text{OH}^-]$  from the concentration, then use  $[\text{H}^+] = K_w/[\text{OH}^-]$ .
- **weak acid** 弱酸: only partly ionised, so use  $[\text{H}^+] = \sqrt{K_a \times [\text{HA}]}$ .

### Buffer solutions

A **buffer solution** 缓冲溶液 resists a change in pH when a small amount of acid or alkali is added. You make one from a weak acid and its conjugate base (for example ethanoic acid and sodium ethanoate).

It works because the mixture holds a store of both partners:

- added  $\text{H}^+$  is removed by the conjugate base:  $\text{CH}_3\text{COO}^- + \text{H}^+ \rightarrow \text{CH}_3\text{COOH}$ .
- added  $\text{OH}^-$  is removed by the weak acid:  $\text{CH}_3\text{COOH} + \text{OH}^- \rightarrow \text{CH}_3\text{COO}^- + \text{H}_2\text{O}$ .

To find the pH, put the concentrations of the acid and its salt into the  $K_a$  expression. Buffers are important in living things—for example,  $\text{HCO}_3^-$  keeps the pH of blood close to 7.4.

### Solubility product

For a salt that barely dissolves, the **solubility product** 溶度积 ( $K_{\text{sp}}$ ) is the product of the ion concentrations in a saturated solution, each raised to the power of its number in the formula:

$$K_{\text{sp}} = [\text{Ag}^+][\text{Cl}^-] \quad K_{\text{sp}} = [\text{Ca}^{2+}][\text{F}^-]^2$$

You can find  $K_{\text{sp}}$  from the solubility, or the solubility from  $K_{\text{sp}}$ .

## The common ion effect

The **common ion effect** 同离子效应 is the way a salt becomes **less** soluble in a solution that already contains one of its ions. The extra ion pushes the dissolving equilibrium back (Le Chatelier), so less salt dissolves. You can calculate the new solubility using  $K_{sp}$  and the concentration of the common ion.

## Partition coefficients

When a solute is shaken with two solvents that do not mix, it spreads between them. The **partition coefficient** 分配系数 ( $K_{pc}$ ) is the ratio of its concentrations in the two layers (at constant temperature):

$$K_{pc} = \frac{[\text{solute in solvent 1}]}{[\text{solute in solvent 2}]}$$

This works when the **solute** 溶质 is in the same physical state in both solvents. The value depends on the **polarity** 极性 of the solute and of each **solvent** 溶剂: a non-polar solute dissolves more in the non-polar solvent, while a polar solute prefers the polar solvent.